

Reimagining Data Workflows with Weaviate Agents

Meet our speakers:



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Agenda

1 An Overview of Weaviate Agents

by Charles Pierse

2 Query Agent

by Tuana Çelik

3 Transformation Agent

by JP Hwang

4 Personalization Agent

by Sebastian Witalec



Weaviate Agents

Let's discuss

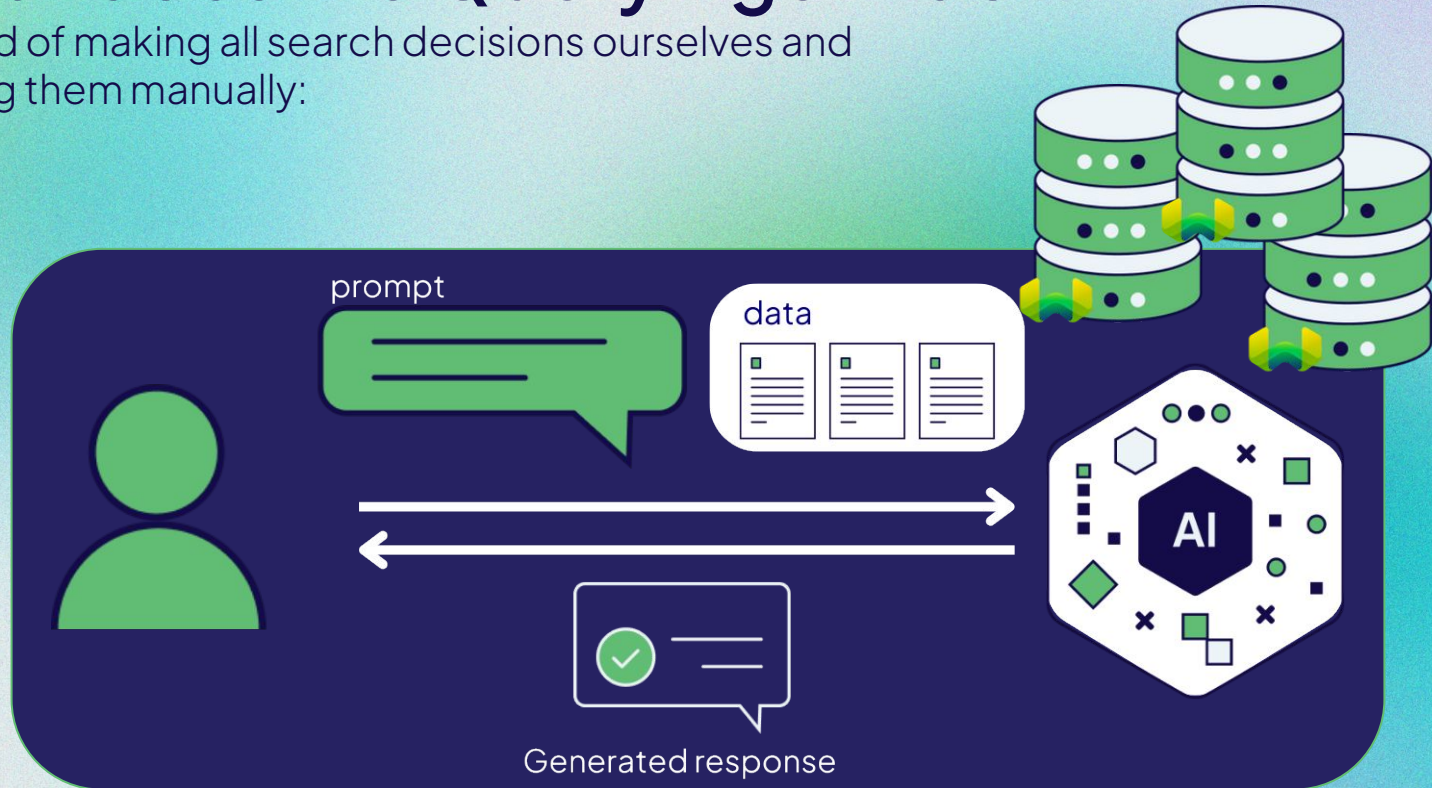


Query Agent

Ask complex questions about your data in Weaviate

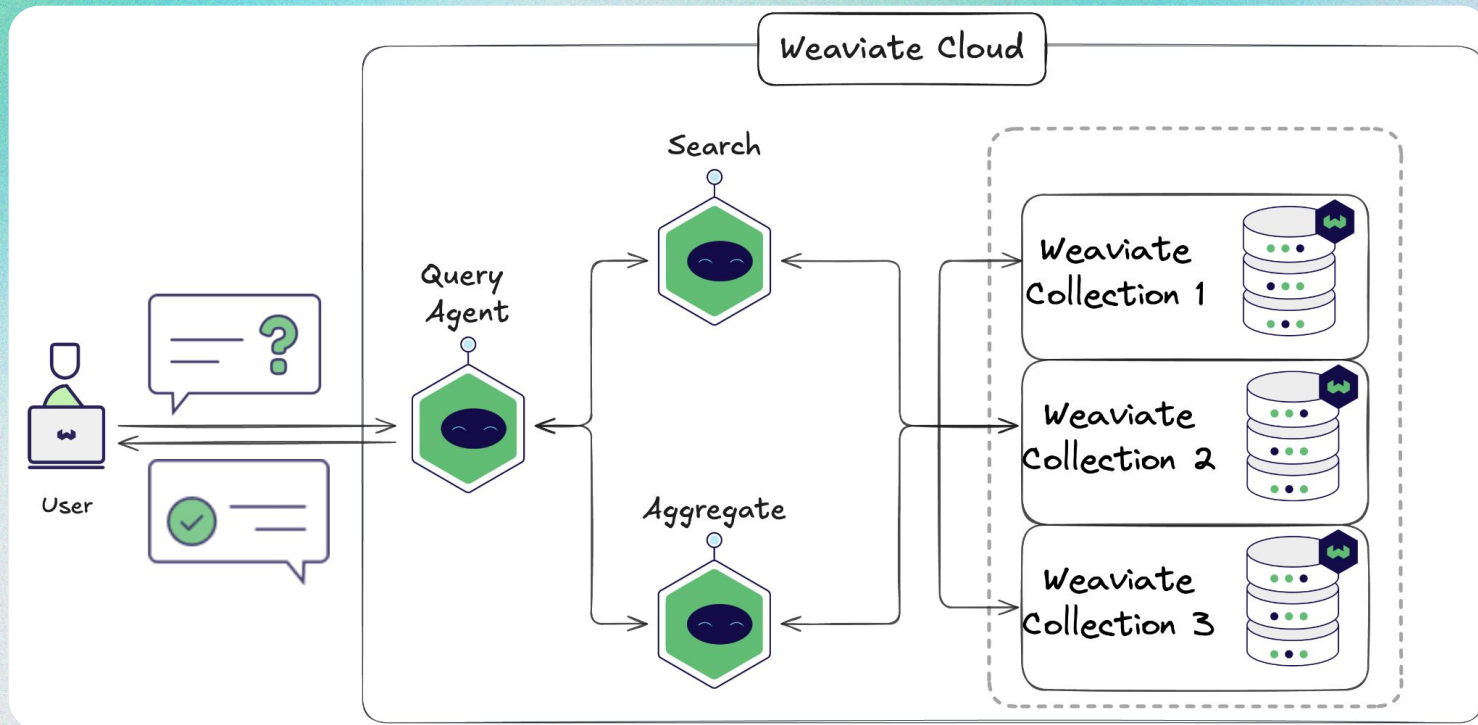
What does the Query Agent do?

Instead of making all search decisions ourselves and running them manually:



What does the Query Agent do?

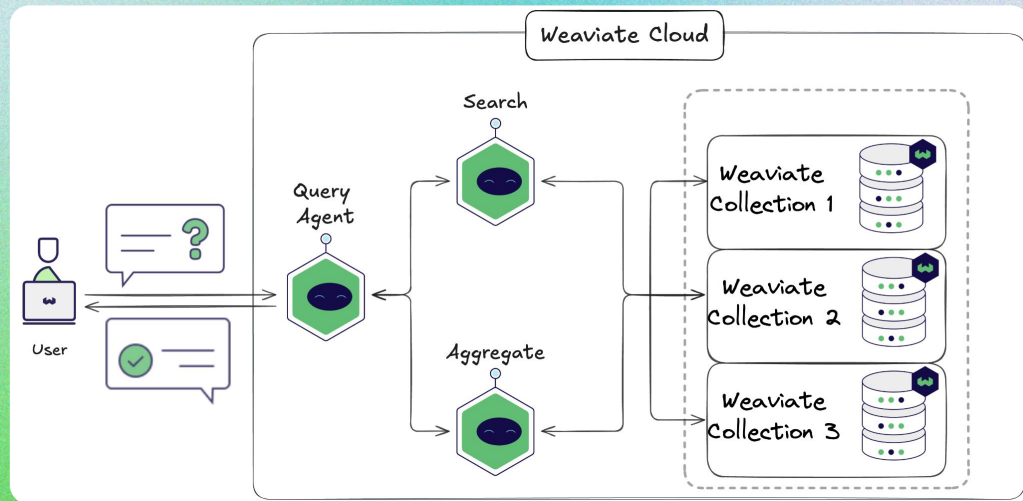
Instead of making all search decisions ourselves and running them manually:



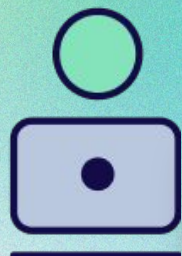
What does the Query Agent do?

Based on a user's query, the agent:

1. Decides which collection(s) to access.
2. Determines if we need regular semantic search, aggregations, or both.
3. Formulates the Weaviate query for each search that will be made.
4. Determines the filters for each query if necessary.
5. Based on the results from all searches, formulates a final answer.



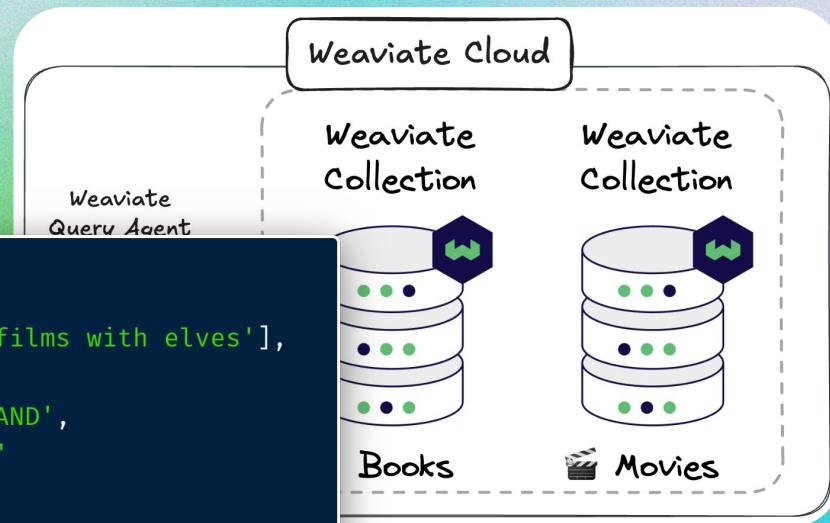
Classifying User Queries



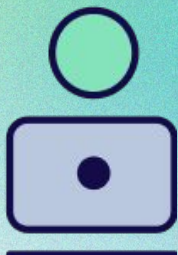
What are some good fantasy films that involve elves?

○ ○ ○

```
QueryResultWithCollection(queries=['fantasy films with elves'],  
                           filters=[[]],  
                           filter_operators='AND',  
                           collection='Movies'  
)
```



Classifying User Queries



Which author has the most books in the 'fantasy'

○○○

```
AggregationResultWithCollection(search_query=None,  
    groupby_property='author',  
    aggregations=[  
        TextPropertyAggregation(  
            property_name='title',  
            metrics=<TextMetrics.TOP_OCCURRENCES: 'TOP_OCCURRENCES'>,  
            top_occurrences_limit=1  
        )],  
    filters=[  
        TextPropertyFilter(  
            property_name='description',  
            operator=<ComparisonOperator.LIKE: 'LIKE'>,  
            value='*fantasy*')],  
    collection='Books'  
)
```

Weaviate Cloud

Weaviate
Collection



Weaviate
Collection

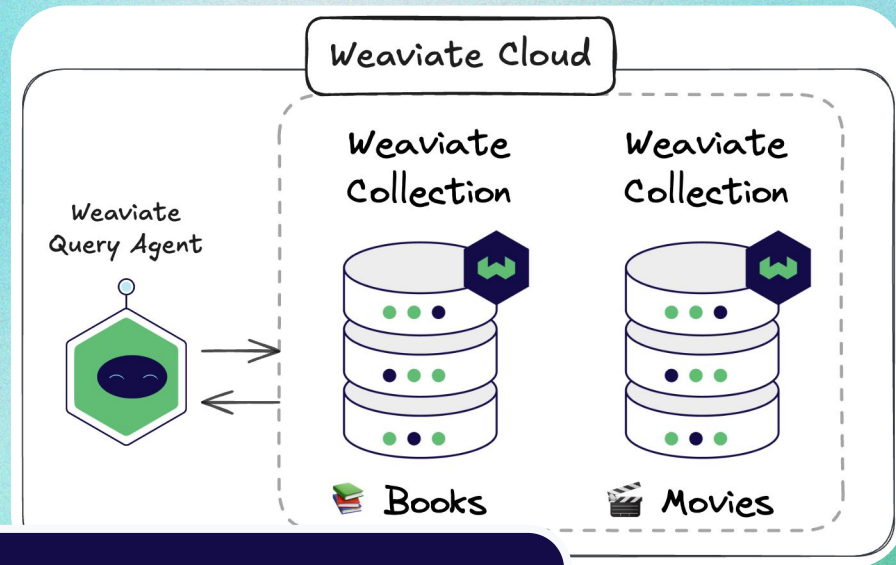


cs



Movies

Classifying User Queries



Demo time!

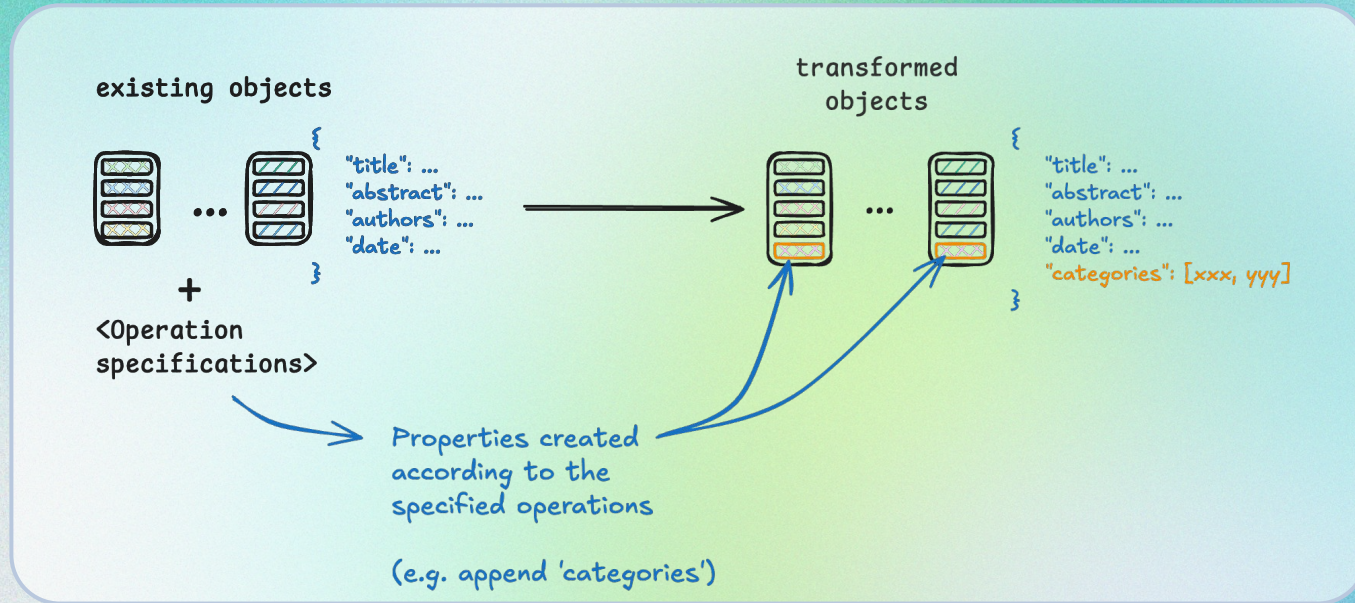


Transformation Agent

Get more out of your data

What does the Transformation Agent do?

It *augments* and *transforms* data



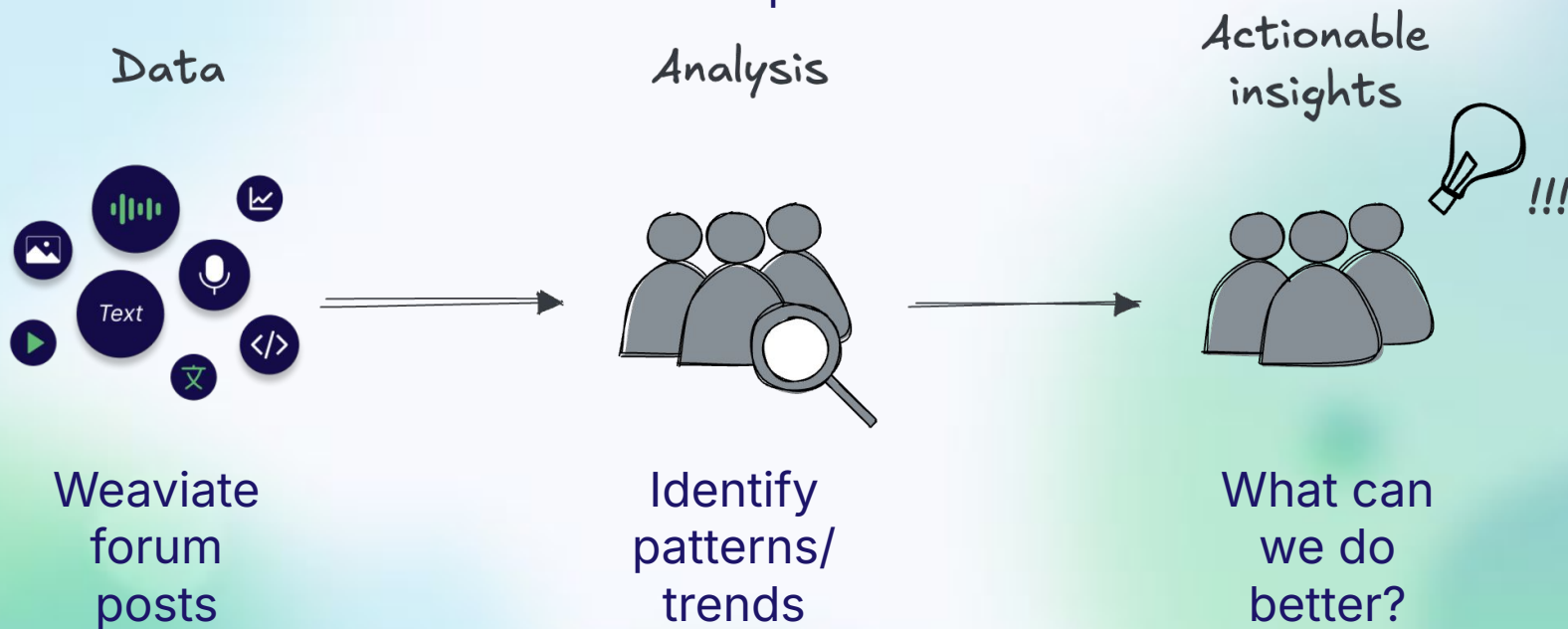
Why use the Transformation Agent?

Typical data workflow



Why use the Transformation Agent?

Example





Why use the Transformation Agent?

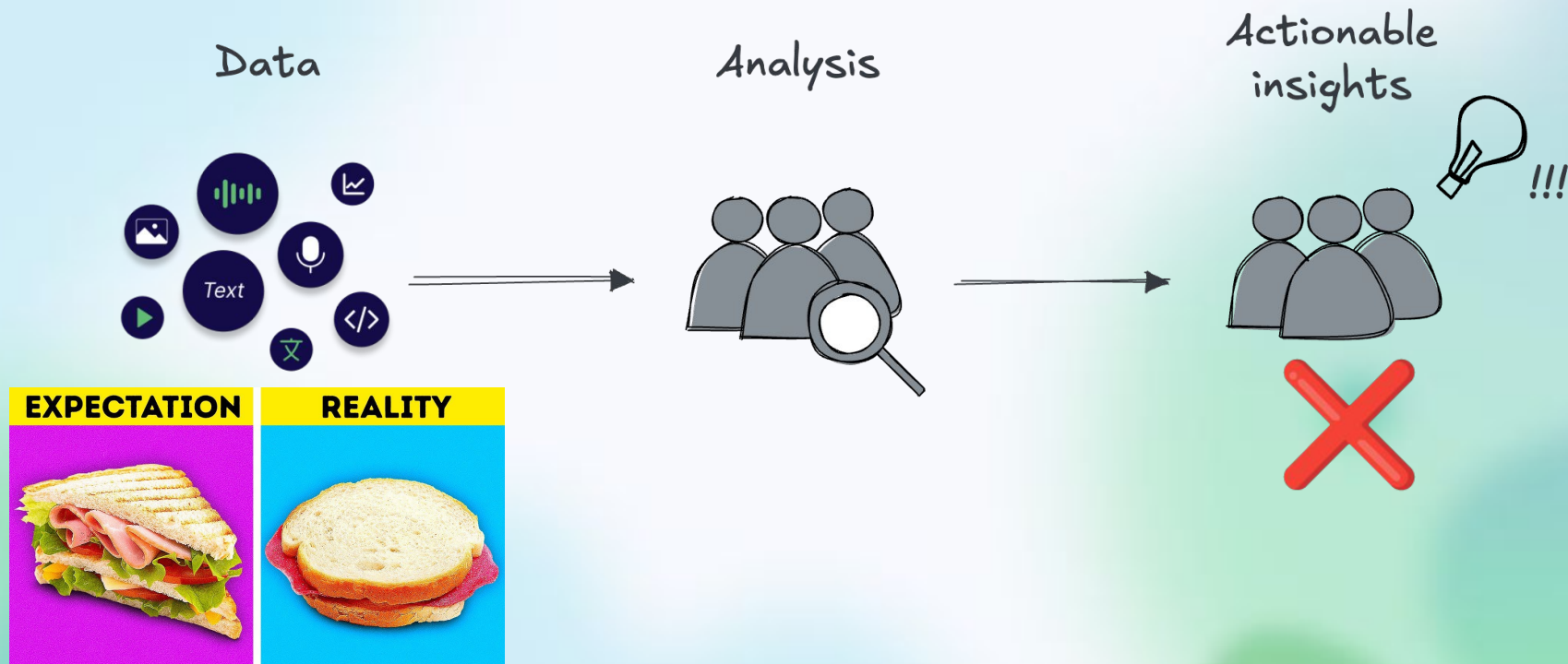
Example forum post

Unstructured text:

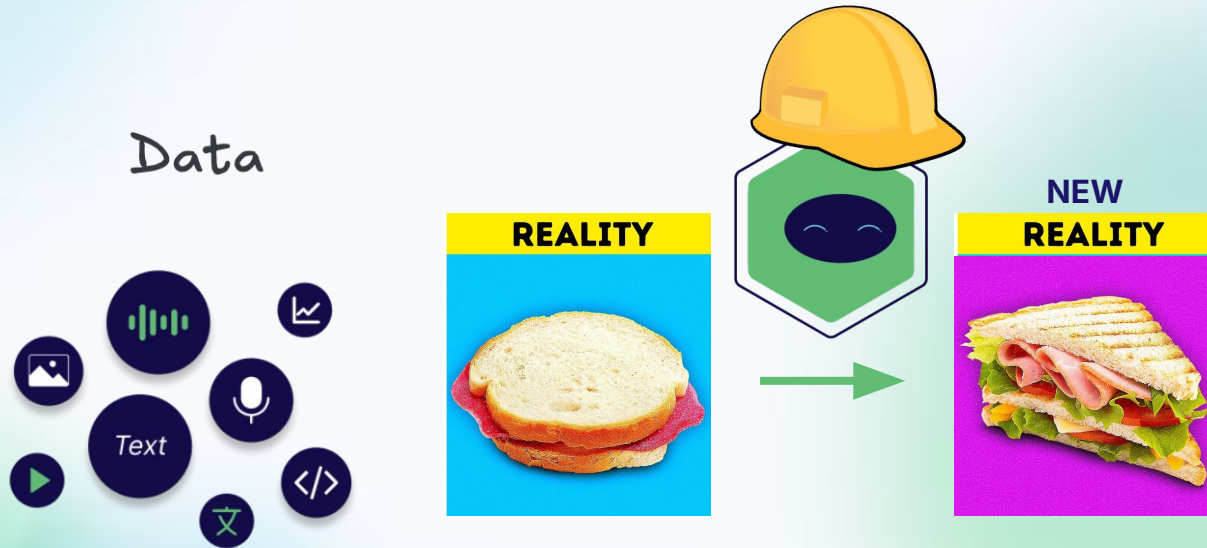
Difficult to manipulate
(group, filter, aggregate, etc)

```
{
  "user_id": 427,
  "conversation": "[Sandip (2024-11-08T07:02:03.772Z)]: I'm
trying to index and search articles in Weaviate, but I'm
encountering an error when querying with near_vector. Here's my
process:\n\nIndexing: I successfully index articles in the...",
  "date_created": "2024-11-08T07:02:03.706Z",
  "has_accepted_answer": true,
  "title": "No such prop with name 'title' found in class
'Test' in the schema\" when querying Weaviate with near_vector",
  "topic_id": 7498
},
```

Why use the Transformation Agent?

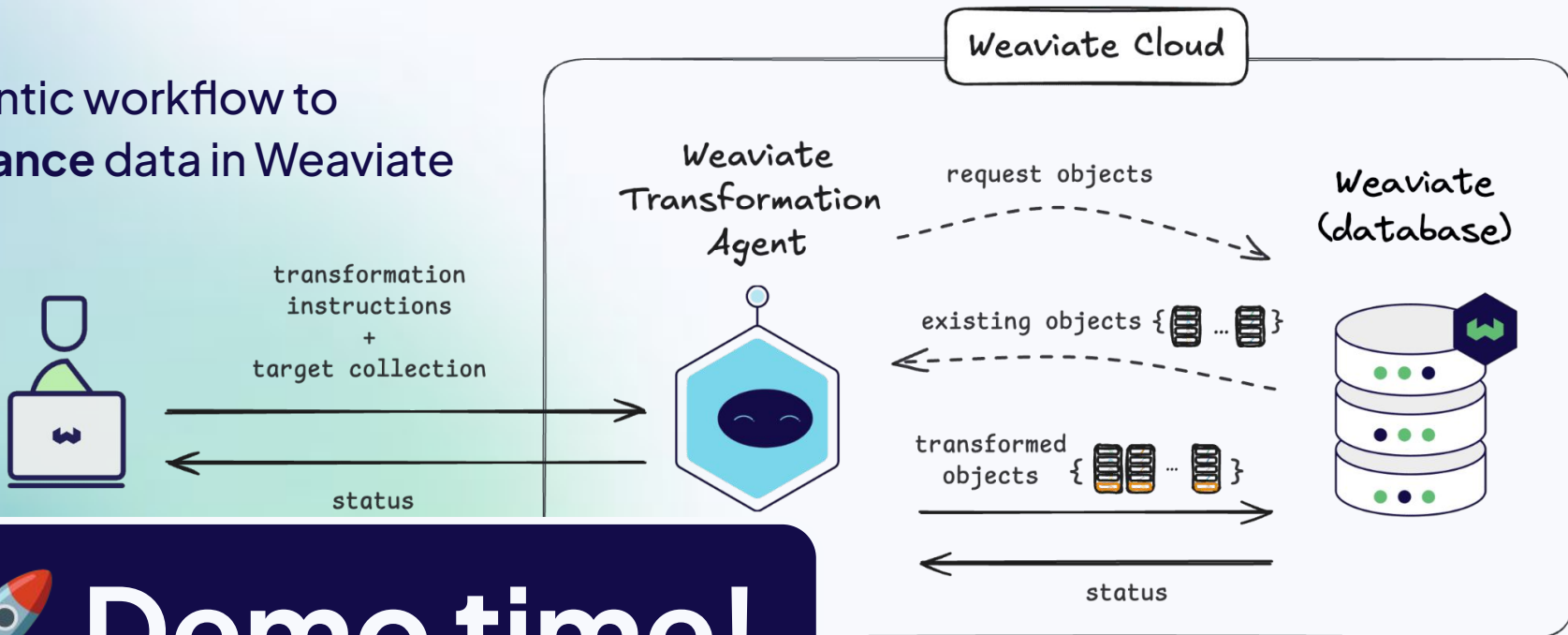


Why use the Transformation Agent?



How does the Transformation Agent work?

Agentic workflow to
enhance data in Weaviate



*Update or add object
properties*



Demo time!

Before transformation

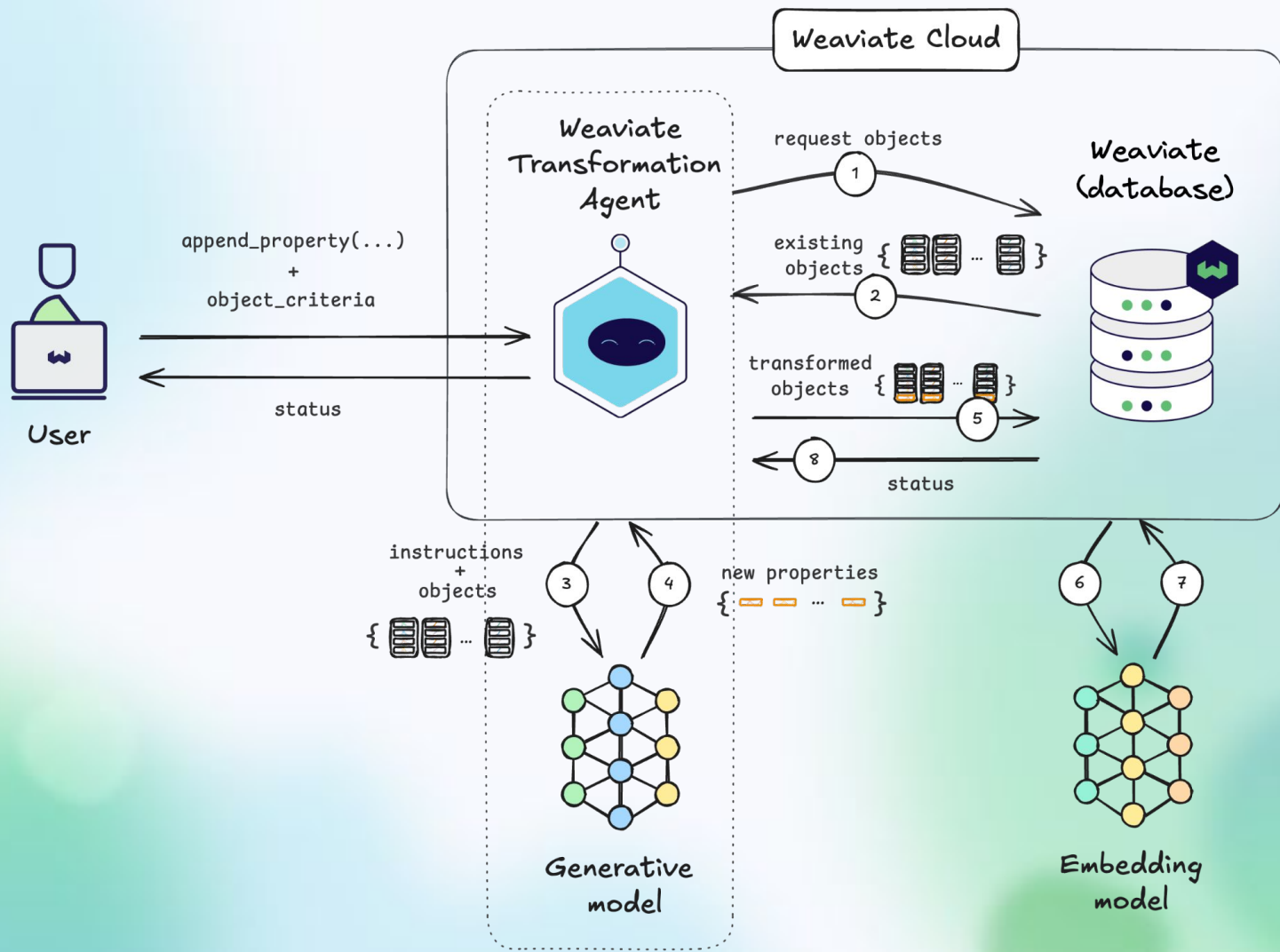
```
{
  "user_id": 427,
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(2024-11-08T07:02:03.772Z)]: I'm trying
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  "title": "No such prop with name
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},
```

Weaviate
Transformation
Agent

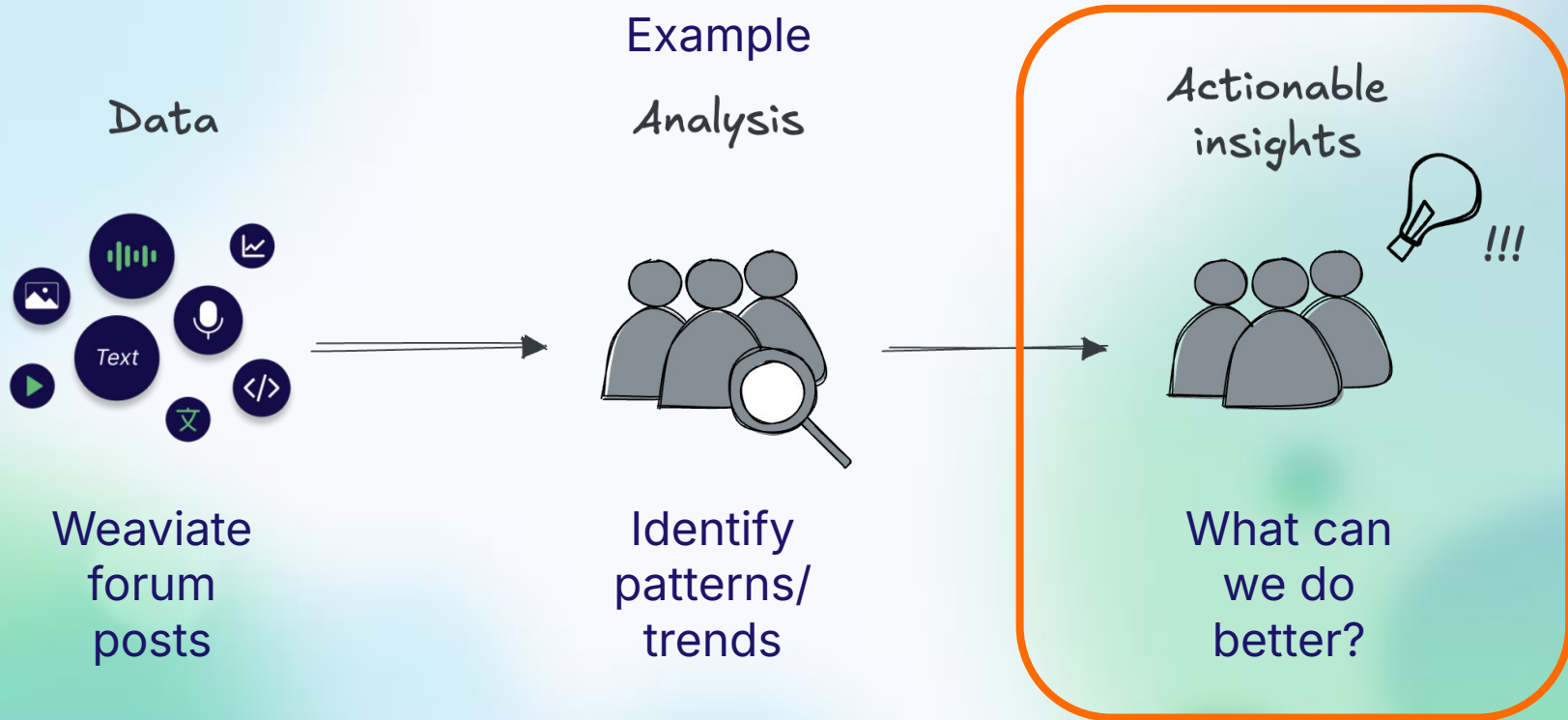


After transformation

```
{
  "user_id": 427,
  "conversation": "[Sandip
(2024-11-08T07:02:03.772Z)]: I'm trying
to index and search articles ...",
  "date_created":
"2024-11-08T07:02:03.706Z",
  "has_accepted_answer": true,
  "title": "No such prop with name
'title' found in class 'Test' ...",
  "topic_id": 7498,
  "accessContext": python_client,
  "technicalComplexity": 3,
  "has_accepted_answer": True,
  "rootCauseCategory":
conceptual_misunderstanding
  "technicalDomain": integration
  "causedByOutdatedStack": False
  "isDocumentationGap": False
},
```



Why use the Transformation Agent?



Actionable insights

How can we help?

Count of Weaviate forum posts, by root cause & technical area.

Root cause category	Deployment	Ingestion	Integration	Queries	Security	Server setup	Others
Conceptual misunderstanding	19	22	31	64	1	12	6
Data modeling	0	1	2	4	0	0	0
Incorrect configuration	43	6	20	11	4	14	0
Incorrect usage	3	4	3	5	0	2	1
Performance	15	21	3	20	0	10	0

Categorized from the raw data by the Weaviate Transformation Agent

Source: Weaviate Forum • Created with Datawrapper

Deeper dive with TA + RAG

```
1 response = collection.generate.fetch_objects(  
2     filters=(  
3         Filter.by_property(name="rootCauseCategory").equal("conceptual_misunderstanding") &  
4         Filter.by_property(name="technicalDomain").equal("queries")  
5     ),  
6     limit=100,  
7     generative_provider=GenerativeConfig.anthropic(model="claude-3-7-sonnet-latest"),  
8     grouped_task="""  
9     From these Weaviate Forum post conversations, identify 3-5 most common things  
10    that we can help users to understand better about Weaviate queries.  
11    If possible, also provide a count of each type in the sample.  
12    """,  
13    grouped_properties=["summary", "title"]  
14 )
```

Actionable insights

Deep-dive into common user misconceptions

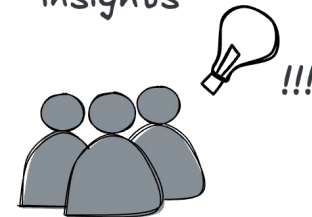
Common re-occurring themes in posts related to "queries" that involved a conceptual misunderstanding.

Understanding Search Types & Behavior	12
Filtering Mechanisms	10
Vector Handling and Configuration	8
Result Scoring and Ranking	7
Query Syntax and Migration	7

Categorized from the raw data by the Weaviate Transformation Agent, and extended

Source: Weaviate Forum • Created with Datawrapper

Actionable
insights



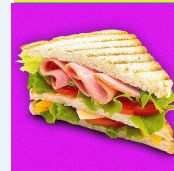
Enabled by:



REALITY

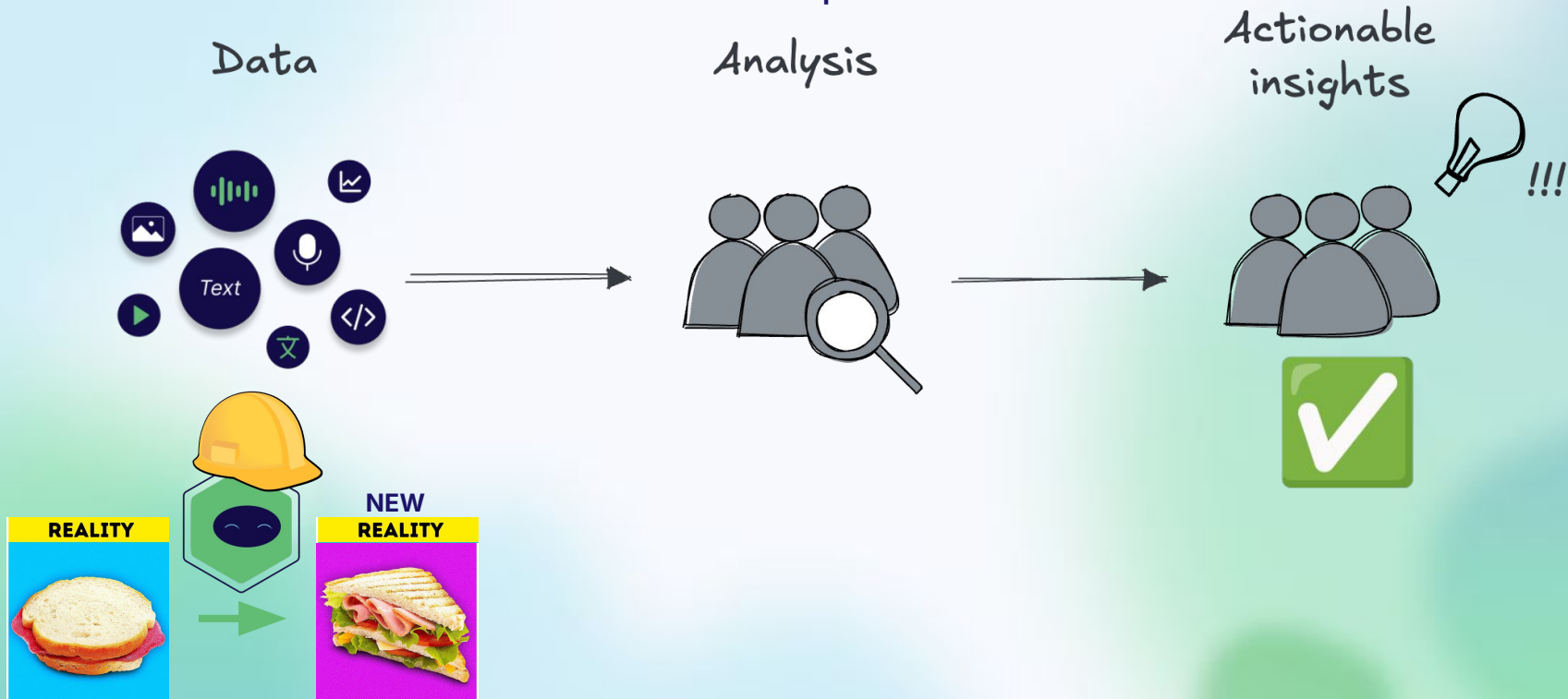


NEW
REALITY



Why use the Transformation Agent?

Example



Simple, natural language instructions

```
1  add_technical_domain = Operations.append_property(  
2      property_name="technicalDomain",  
3      data_type=DataType.TEXT,  
4      view_properties=["conversation", "title"],  
5      instruction=f"""  
6          Identify the primary technical domain of the user's forum post query.  
7          The answer must be one of the following:  
8          {TECHNICAL_DOMAIN_CATEGORIES.keys()}  
9  
10         The definitions of the categories are as follows:  
11         {TECHNICAL_DOMAIN_CATEGORIES}  
12  
13         Remember that the answer must be one of these categories:  
14         {TECHNICAL_DOMAIN_CATEGORIES.keys()}  
15         """,  
16  )
```

Weaviate
Transformation
Agent



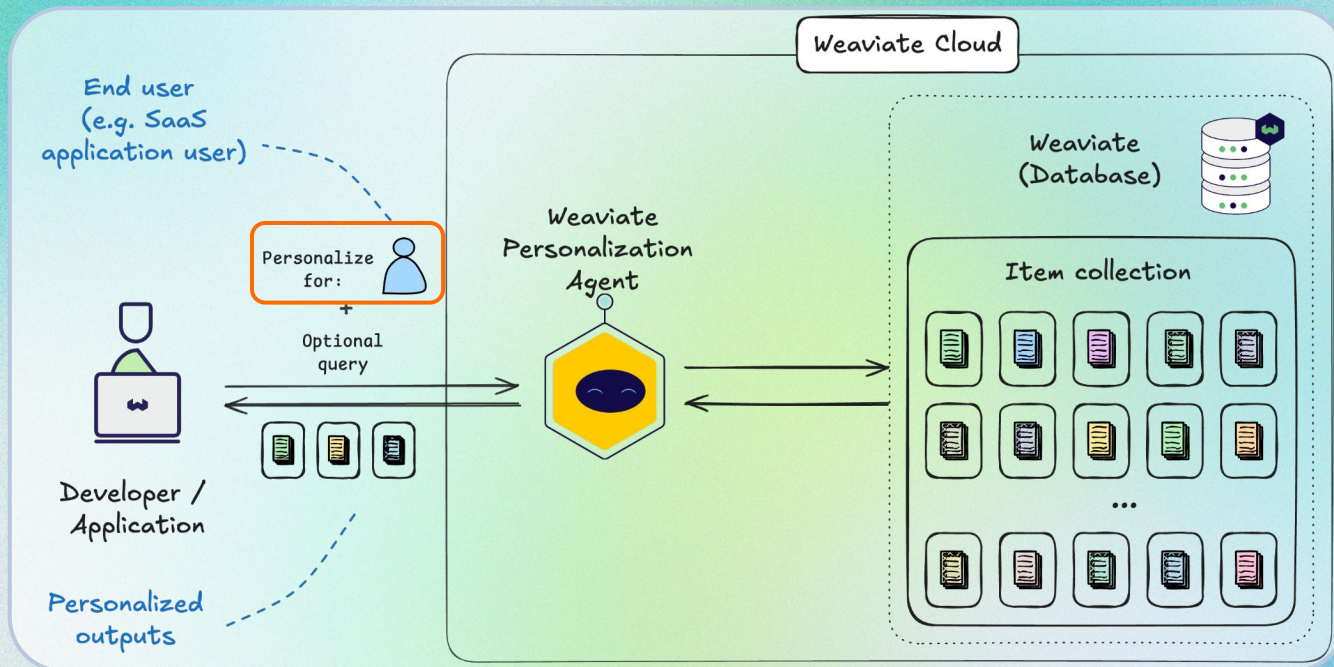


Personalization Agent

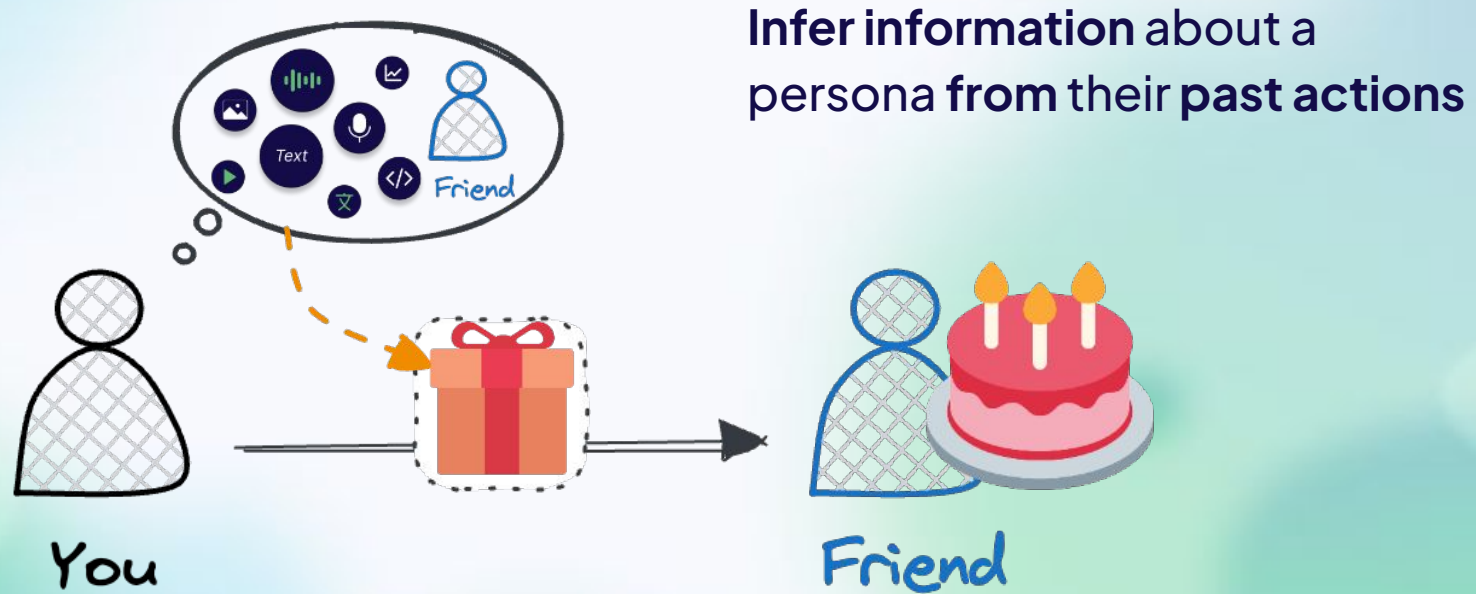
Data matchmaking

What does the Personalization Agent do?

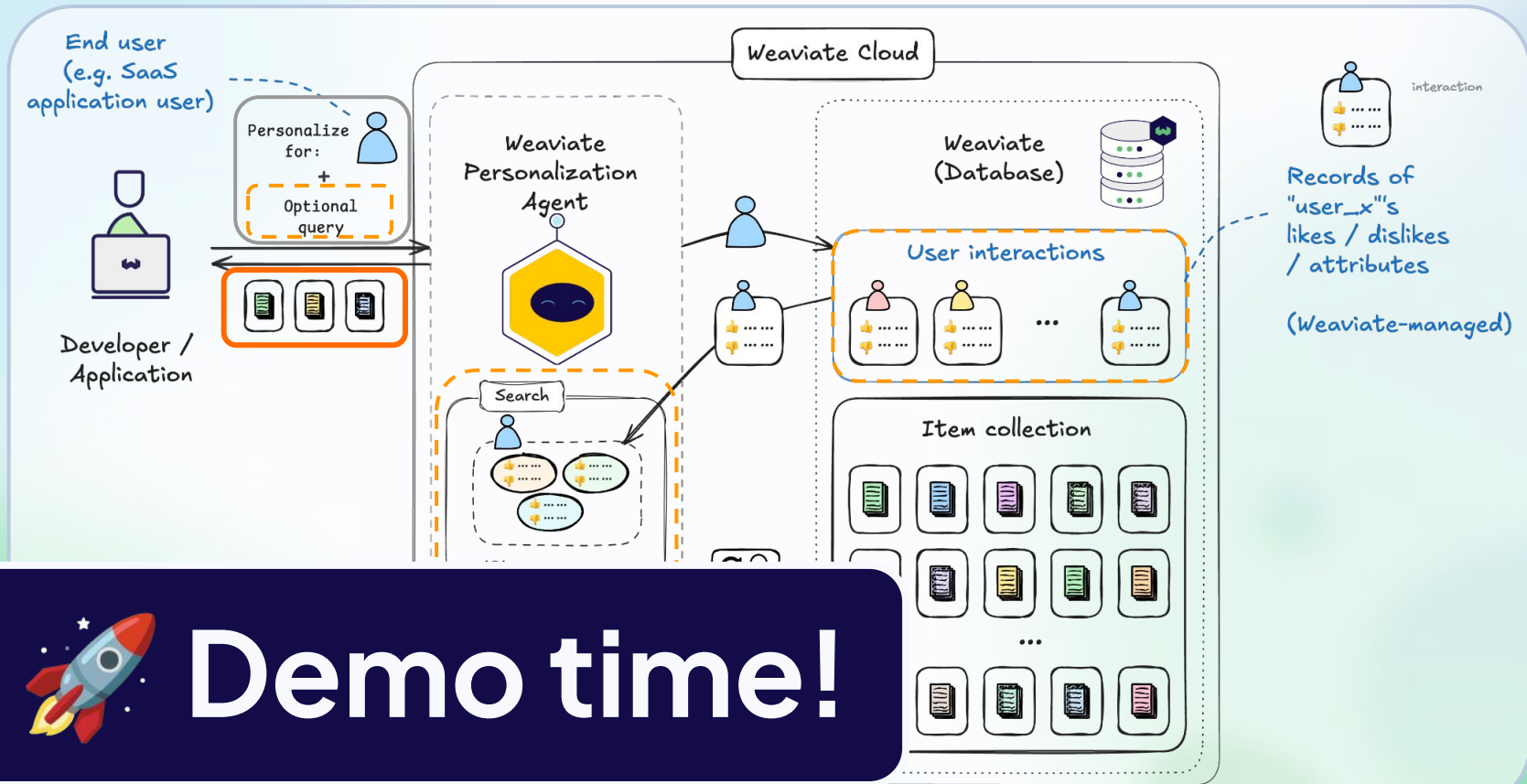
It returns *personalized recommendations* for each user



Why use the Personalization Agent?



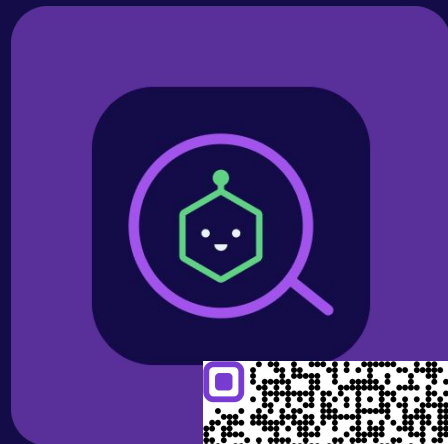
How does the Personalization Agent work?



Demo time!

Q&A

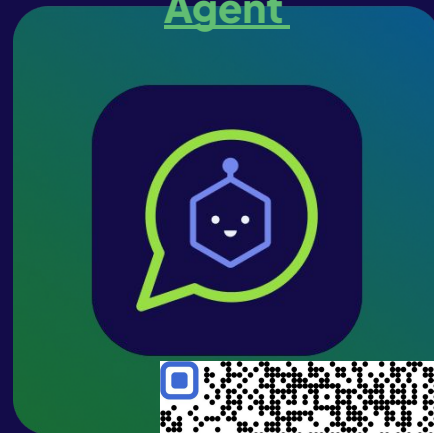
Query Agent



Transformation Agent



Personalization Agent



**THANK
YOU**



LOVED THIS SESSION - SIGNUP FOR FOLLOW UP MATERIAL